Luca Panconi, PhD., MSci.

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SUMMARY

- **Computational biologist**, with expertise in **machine learning and computer vision.**
- **8+ years experience** in software development with **Python and R**.
- Currently employed at **the Swedish AI Factory** as an **AI Application Specialist**.
- Strong command of **machine and deep learning**, **data science** and **computer vision for biomedical imaging**.
- Previous **RED Postdoctoral Fellowship** holder at **SciLifeLab**, with research focus on **data-driven ML** for **computationally-optimised immunotherapy**.
- **First author of five research publications** from research undertaken during academic career.
- **Experienced scientific communicator**, presenting at 10+ academic conferences.

EMPLOYMENT HISTORY

January 2026 – Present

AI Application Specialist, MIMER (the Swedish AI Factory)

- Main responsibilities include **designing scalable software solutions**, developing **production-grade ML pipelines**, and deploying models on **high-performance computing infrastructure**.
- Served as interim **Team Coordinator** for the **Life Science Technical Team.**
- Provided **technical expertise** through AI consultancy to over 50 **academic and industrial collaborators**.
- Covered all facets of software development, including **prototyping**, **production-level coding, technical evaluation,** and **deployment.**
- Ensuring **responsible, trustworthy and ethical use** of AI, maintaining **DevOps** and **MLOps** to the highest standard.

September 2024 – January 2026

RED Postdoctoral Research Fellow, SciLifeLab, Stockholm University

- Awarded the Research, Environment and Development (RED) **postdoctoral research fellowship**.
- Main research project centred around developing **AI-enhanced microscopy and image analysis techniques** for **personalised medicine**.
- Responsibilities include project leadership, supervising PhD and Masters students, presenting seminars, grant writing and managing collaborations.

April 2024 – July 2024

Postdoctoral Research Fellow, Institute of Immunology and Immmunotherapy, University of Birmingham

- Developing AI for analysis of **membrane protein dynamics.**
- **Mentoring** graduate students, providing guidance on academic development.

EDUCATION

September 2020 – September 2024
Topological Design PhD, University of Birmingham, UK

September 2016 – June 2020
Mathematics MSci (First Class Honours, Magna Cum Laude), University of Birmingham, UK

NOTABLE PUBLICATIONS

AI4CellFate: Interpretable early cell fate prediction with generative AI
I. Cunha, **L. Panconi**, S. Bauer, M. Gestin, E. Latron, E. Sahai, A. Le Marois, J. Griffié.
bioRxiv, 2025. DOI: https://doi.org/10.1101/2025.05.12.653464.
Accepted in *Nature Communications*.

SMLMFlow: Improving structural resolution in single molecule localization microscopy with flow matching
S. Bauer, **L. Panconi**, I. Cunha, E. Latron, D. Sage, R. Peters, J. Griffié.
bioRxiv, 2025. DOI: https://doi.org/10.1101/2025.05.12.653464.
Under review with *Nature Communications*.

Mapping membrane biophysical nano-environments
L. Panconi, J. Euchner, M. Makarova, D-P. Herten, D.M. Owen, D. J. Nieves.
Nature Communications 15 (1), 2024. DOI: https://doi.org/10.1038/s41467-024-53883-1

Three-dimensional topology-based analysis segments volumetric and spatiotemporal fluorescence microscopy
L. Panconi, A. Tansell, A. J. Collins, M. Makarova, D.M. Owen.
Biological Imaging 3 (1), 2024. DOI: https://doi.org/10.1017/S2633903X23000260.

Methods for imaging cell membranes (1st ed.)
L. Panconi, D. Nieves, M. Makarova, D. M. Owen.
CRC Press, 2023. DOI: https://doi.org/10.1201/9781003273745.

Cluster analysis for localisation-based data sets: dos and don'ts when quantifying protein aggregates
L. Panconi, D. M. Owen, J. Griffié.
Frontiers in Bioinformatics 3 (1), 2023. DOI: https://doi.org/10.3389/fbinf.2023.1237551.

Phospholipid tail asymmetry allows cellular adaptation to anoxic environments
L. Panconi, C. D. Lorenz, R. C. May, D. M. Owen, M. Makarova.
Journal of Biological Chemistry 299 (9), 2023. DOI: https://doi.org/10.1016/j.jbc.2023.105134.

Topology-based fluorescence image analysis for automated cell identification and segmentation
L. Panconi, M. Makarova, E. R. Lambert, R. C. May, D. M. Owen.
Journal of Biophotonics 16 (3), 2023. DOI: https://doi.org/10.1002/jbio.202200199.

Nanoscale protein clustering modulates the input-output response of cellular signalling pathways
L. Panconi, J. Griffié, D.M. Owen.
bioRxiv, 2025. DOI: https://doi.org/10.1101/2025.03.13.642965.
Accepted in Scientific Reports.

A generalised molecular dynamics simulator for tracking transmembrane protein aggregation

L. Panconi, O. Dalby, C. Zaza, D.M. Owen, S. Simoncelli, J. Griffié.
Manuscript in preparation.

PUBLICATION REVIEWS

A novel morphological descriptor for multiphase microstructure reconstruction

Dong Dong Chen, Jiao Fen Nan, Xiao Rui Wang, Xiang Ru Chen, Bin Jiang
Journal of Microscopy 301 (3), 2026. DOI: https://doi.org/10.1111/jmi.70052.

The method for assessing the specificity of developing CAR therapies

Ivan V. Prikhodko, Georgy T. Guria.
Biophysical Reports 4 (3), 2024. DOI: https://doi.org/10.1016/j.bpr.2024.100172.

CONFERENCE TALKS

September 2024

5th Annual SciLifeLab Science Talks

- Presented an **invited talk** on novel agent-based modelling approaches for studying TCR signal transduction and informing therapeutic choices.

June 2024

European Light Microscopy Initiative (ELMI)

- Presented a talk on topological data analysis techniques for **identifying nanoscale lipid domains** from super-resolution microscopy data.

August 2022

University of the Third Age (U3a)

- **Gave a seminar** on super-resolution microscopy, lipid order and methods of analysing marked point patterns to the U3A group in Birmingham, UK.

July 2022

Frontiers in Bioimaging

- **Presented a talk** on **topological image analysis**, with an emphasis on using Topological Boundary Line Estimation using Recurrence of Neighbouring Emissions (TOBLERONE) for **cell segmentation**.

July 2022

University of Birmingham Mathematics Postgraduate Society Poster Session

- **Poster presentation** on the **applications of topology to image analysis**.

July 2022

British Early Career Mathematicians' Colloquium

- **Presented a talk** on topological data analysis techniques for **bioimaging**.

June 2022

University of Birmingham Postgraduate Research Festival

- **Poster presentation** on the applications of topological data analysis in **quantitative biology**.

June 2021
Young Researchers in Mathematics
- **Presented a talk** on the applications of mathematical modelling to **biophysical and thermodynamic systems**.

February 2018
Tomorrow's Mathematicians Today
- **Presented a talk** on the modern applications of Boolean algebra, particularly in regards to **data science and programming**.

POSTER PRESENTATIONS

November 2024
Data-Driven Life Science Annual Conference
- **Poster presentation** on applications of **machine learning** to modelling **transmembrane protein dynamics**.

August 2023
Single Molecule Localisation Microscopy Symposium
- **Poster presentation** for applications of **pattern segmentation techniques** to artificial and whole cell membrane data.

April 2023
Focus on Microscopy
- **Poster presentation** on marked point clustering and colocalisation analysis techniques, with an emphasis on **super-resolution data processing**.

November 2022
Institute of Immunology and Immunotherapy Away Day
- **Poster presentation** on **analysis techniques for marked point patterns** as part of the Point Label Analysis for Super-resolved Membrane Attributes (PLASMA) project.

POSITIONS OF RESPONSIBILITY

September 2022 – September 2024
CDT in Topological Design Outreach and Public Engagement Committee, University of Birmingham
- **Organised workshops and lectures** based on the applications of mathematics and topology.
- **Delivered talks to over two hundred students** aged 14-18 across disadvantaged regions of Birmingham, UK.
- Designed **committee website** and distributed online resources.

September 2021 – September 2022
Applied Topology Seminar Committee, University of Birmingham
- Worked as **Liaison Officer** for the Applied Topology seminar series.

- **Coordinated and hosted a series of seminars** with both local and international speakers.

September 2020 – September 2021

Applied Mathematics Seminar Committee, University of Birmingham

- Organised Applied Mathematics seminar series, including venues. **Hosted and introduced speakers**.
- Sourced potential speakers both **locally and globally.**

INDUSTRY PLACEMENTS

April 2023 – July 2023

Oxford Nanoimaging (ONI), Oxford, UK

- Worked in **back-end development** with the ONI software team.
- **Implemented machine learning techniques** for microscopy data, which were adopted into their online analysis platform.

TEACHING EXPERIENCE

January 2021 – September 2025

Teaching Assistant, University of Birmingham, UK

- Worked as a **Post-graduate Teaching Assistant** (PGTA) alongside PhD studies.
- Collaborated with module leads to **plan courses for undergraduate students** and grade coursework.
- **Worked with lecturers nationally and globally,** from both my home University as well as the University of Birmingham's institute in Jinan, China.

August 2022 – September 2024

Stem Ambassador, Stem Learning

- **Organised outreach** activities to local schools and colleges.
- Events are run throughout the year, aimed at giving **students aged 14-18** an insight into university-level research.
- **Co-ordinated and delivered lectures** on applications of topology as part of the CDT in Topological Design's Outreach scheme.

September 2022 – December 2022

Essentials of Mathematics, Statistics and Programming, University of Birmingham

- Worked as a teaching assistant for a course on the fundamentals of **mathematics, bioinformatics and programming**.
- **Gave two 3-hour lectures** on data processing and visualisation.

January 2022 – April 2022

Mathematical Modelling and Problem Solving, University of Birmingham

- Worked as a teaching assisting on a **mathematical modelling and problem solving** module.
- **Delivered two lectures to a cohort of over 100 undergraduate students** on methods of presenting mathematical research.

January 2022 – April 2022

Undergraduate Supervision, University of Birmingham

- **Supervised two undergraduate students,** in their third year of study, undertaking a research project based on SMLM and data analysis.
- Gave mentorship in both **wet lab work and programming**.
- **Prepared teaching materials** detailing lab protocols.

ADDITIONAL FUNDING AND AWARDS

July 2022
Best Poster Prize, University of Birmingham Mathematics Postgraduate Society
- Awarded for work on topological image analysis and applications to **cell segmentation and bioinformatics**.

June 2022
College of Medical and Dental Sciences Special Award, University of Birmingham Graduate School
- Awarded to the **best poster presentation** within the College of Medical and Dental Sciences. Based on topological image analysis in bioimaging.

March 2021
HEFI Horizon Award, University of Birmingham
- Awarded for going **above expected teaching requirements** and completing additional courses in education.

December 2020
Graham Hoare Prize, Institute of Mathematics and its Applications
- Awarded for **writing an article**: "A Mathematician's Guide to the Perfect Mug of Tea", featured in Mathematics Today.

September 2020
Studentship, Engineering and Physical Sciences Research Council
- Awarded for **studentship to the CDT in Topological Design**. Given as an **annual stipend of £17506**, awarded monthly for the course of four years.

March 2019
B-Enterprising Ideas Fund, University of Birmingham
- Awarded for **producing a successful business application** to set up a small initiative creating mathematics resources for students outside of mathematical disciplines. Received £500 for set up costs and equipment.

November 2016
Personal Skills Award, University of Birmingham
- Awarded for a range of extra-curricular activities including **conservational volunteer work**, **fundraising for charity** and completing courses on select **administrative skills**.

September 2016
Watson Award, University of Birmingham School of Mathematics
- Awarded to **outstanding mathematics students**. Given as a scholarship summing to **£1000**.